MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6X5WGT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, high vacuum, full wave

Outline	---	9-11 or 9-41 (EIA)
Base	---	B6-8, B6-48, or B6-60 (phenolic)
Envelope	---	T9
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	5	7	8
Element	---	nc	h	2a	1a	h	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	epx	Epp/a	ib	Io	i(surge)	Rℓ	Cℓ	Ehk	Alt
Unit:	V	v	Vac	ma	mAdc	ma	Ohms	µF	v	ft
Maximum:	6.9	1,375	---	230	75	750	---	---	450	Note 1
Minimum:	5.7	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	---	400	---	---	---	5,700	8	---	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1353	Operation of rectifiers	2,3		0.65	II	Io	70	---	mAdc
1231	Emission (1)	-	E2b = 0; E1b = 50 Vdc	0.65	II	I1s	140	---	mAdc
1231	Emission (2)	-	E1b = 0; E2b = 50 Vdc	0.65	II	I2s	140	---	mAdc
1201	Short and discontinutiy detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
1301	Heater current	-		---	---	If	540	660	mA
1336	Heater-cathode leakage	-	Ehk = Eo	---	---	Ihk	---	150	µAdc
1041	Shock	-	450 G	15	S2	---	---	---	---
---	Shock and vibration-fatigue test end points:								
1353	Operation of rectifiers	-		---	---	Io	60	---	mAdc
1336	Heater-cathode leakage	-		---	---	Ihk	0	200	µAdc
1101	Secureness of base, cap, or insert	-		---	---	---	---	---	---
1111	Base pin solder depth (rigid leads)	-		---	---	---	---	---	---
1211	Insulation of electrodes	-		4.0	S3	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
1501	Intermittent life-test	4	Group B	---	---	---	---	---	---
---	Intermittent life-test end point (500 hours):								
1353	Operation of rectifiers	-		---	---	Io	60	---	mAdc
1031	Vibration fatigue	5		---	---	---	---	---	---
1031	Variable-frequency vibration	5	No voltages applied	---	---	---	---	---	---

NOTES:

1. See reduced pressure (altitude) rating, and altitude, maximum peak voltage in the basic document.
2. This test to be performed at the conclusion of the holding period.
3. In a full-wave circuit adjust Zp/a such that a tube having Etd = 22 Vdc at 70 mAdc per anode gives Io = 75 mAdc.
4. In the life-test operation the values of Rt and Ct given in test conditions may be considered as approximate and shall be adjusted initially to give Io equal to, or greater than, Io = 75 mAdc with ib equal to, or greater than, 205 ma. Ehk = Eo.
5. (B) This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made on samples of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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Air Force - 17, 80
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User activities:
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3050)

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